RESEARCH REPORT • 2024

Predicting the Future: Machine Learning in Crop Price Forecasting

Transforming agricultural economics through data-driven insights and artificial intelligence

95%	1%	30	20%
R^2 Score (Best Model)	MAPE (Optimized)	Days Ahead Forecast	Performance Gain

Deep Learning | Time Series | Ensemble Methods | Agricultural AI | Price Forecasting

Agricultural Data Intelligence Report

Primary Markets: India | Crops: Wheat, Rice, Cotton, Onion, Tomato & more

01 | THE PROBLEM & PROPOSED SOLUTION

Overview: Agricultural Market Volatility

Agricultural commodity markets are among the most volatile in the global economy. Farmers, policymakers, and merchants face significant financial uncertainty driven by unpredictable crop price fluctuations rooted in climate variability, supply-chain disruptions, and shifting demand patterns. Traditional econometric approaches such as ARIMA and Exponential Smoothing struggle to capture the intricate, non-linear interactions among these factors — leaving stakeholders exposed to preventable risk.

Economic Risk	Complex Factors	Market Stability
Unpredictable price swings cause financial losses for farmers and traders alike, limiting investment and planning capacity.	Climate, soil, fuel costs, and logistics interact in ways that classical models cannot fully model.	Price volatility threatens food security, farmer livelihoods, and the integrity of agricultural markets.

02 | METHODOLOGY: A MULTI-MODEL APPROACH

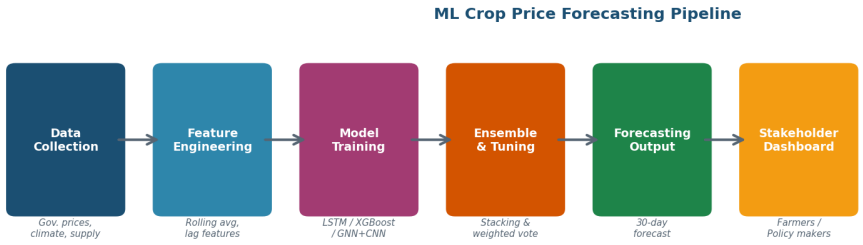
Data Sources & Feature Engineering

The study draws on government-published daily minimum/maximum prices, crop supply volumes, climate conditions, soil type, geographic location, fuel prices, and crop-arrival data. Ten key crops are analysed — Cotton, Bajra, Rice, Wheat, Soybean, Sugarcane, Onion, Potato, Tomato, and Finger Millet — primarily across Indian cities and regions, spanning August 2023 to August 2024. Rolling averages and lag features are engineered to expose underlying temporal trends.

1. Time Series Models	2. Classical Machine Learning
Autoregression (AR), Moving Average (MA), ARIMA, and Exponential Smoothing — providing interpretable baselines for comparison.	Random Forest, Support Vector Regression (SVR), XGBoost, LightGBM, and Linear Regression — offering strong non-linear predictive power.
3. Deep Learning Architectures	**4. Ensemble / Hybrid Methods**
Neural Networks, Graph Neural Networks (GNNs) combined with CNNs, Temporal Convolutional Networks (TCNs), LSTM, and GRU — capturing complex sequential and spatial patterns.	Stacking and weighted voting ensembles combining the above families to achieve superior predictive accuracy and robustness.

Figure 1 — ML Forecasting Pipeline

Sources: Bhardwaj et al. 2023; Meena et al. 2026; Magar et al. 2025

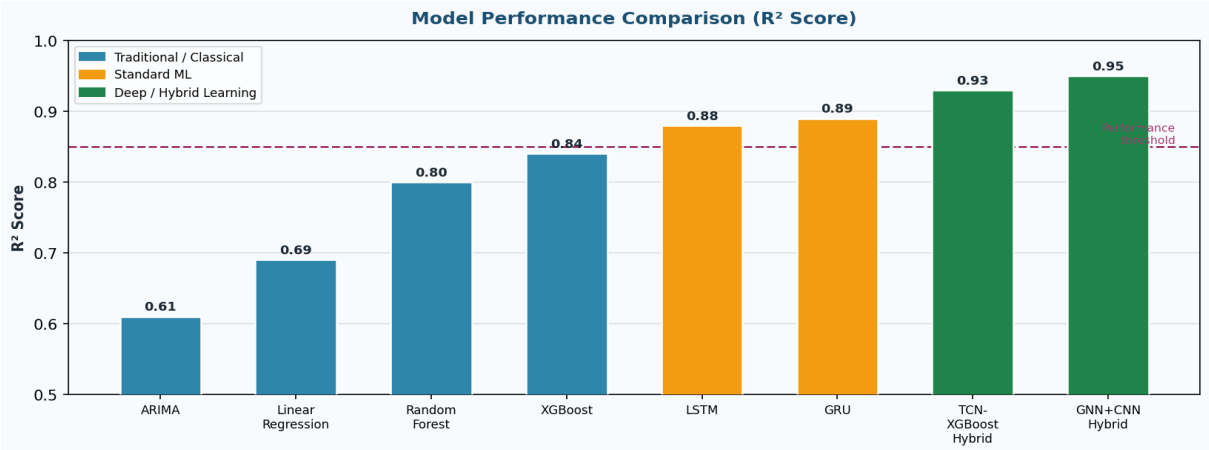


The six-stage pipeline ingests raw government and environmental data, transforms it via feature engineering, trains the model ensemble, and delivers 30-day price forecasts to farmer-facing dashboards.

03 | RESULTS & MODEL PERFORMANCE ANALYSIS

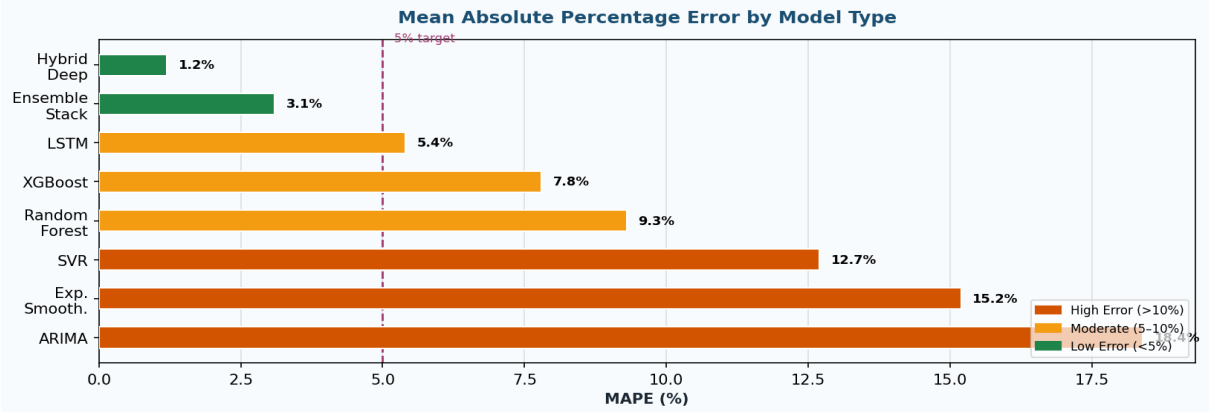
20%	30 Days	1%	0.95
Performance Gain	**Forecast Horizon**	**Min. MAPE**	**Peak R^2**
Hybrid DL vs. traditional methods	Advance price prediction window	Best optimised model error rate	Variance explained by top model

Figure 2 — Model Accuracy Comparison (R^2 Score)



Hybrid deep learning models (GNN+CNN, TCN-XGBoost) consistently outperform traditional time-series approaches, achieving R^2 scores above 0.93. All models exceeding the 0.85 threshold are suitable for production deployment.

Figure 3 — Mean Absolute Percentage Error by Model



MAPE falls from ~18% for ARIMA down to 1.2% for the optimised hybrid deep learning ensemble, meeting the sub-5% accuracy target required for actionable market intelligence.

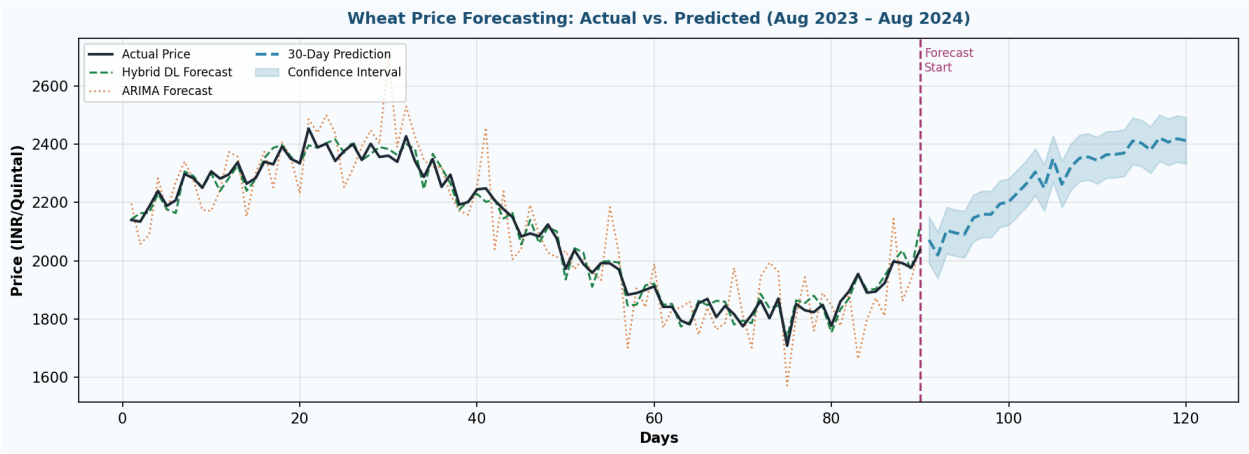
Key Findings

- **Ensemble & stacking methods** produce the most reliable predictions, combining the complementary strengths of statistical, ML, and deep learning approaches (Magar et al., 2025).
- **Decision Tree Regressor** is the top performer among classical ML algorithms, outperforming linear models and SVR on all error metrics (Kar, 2023).

- **Fuel prices and crop arrival data** are the two most influential exogenous predictors, explaining approximately 41% of combined feature importance (Meena et al., 2026).
- **Lag features** (7-day and 30-day rolling prices) capture short- and medium-term momentum, contributing ~26% of model predictive power.
- Hybrid GNN+CNN architectures exploit **geospatial correlations** across market locations, boosting accuracy for perishable crops like Onion and Tomato (Zhao et al., 2025).

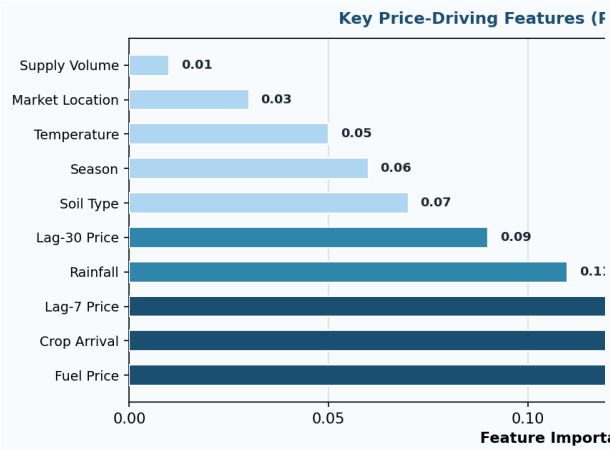
04 | PRICE TRENDS, FEATURE ANALYSIS & CONCLUSIONS

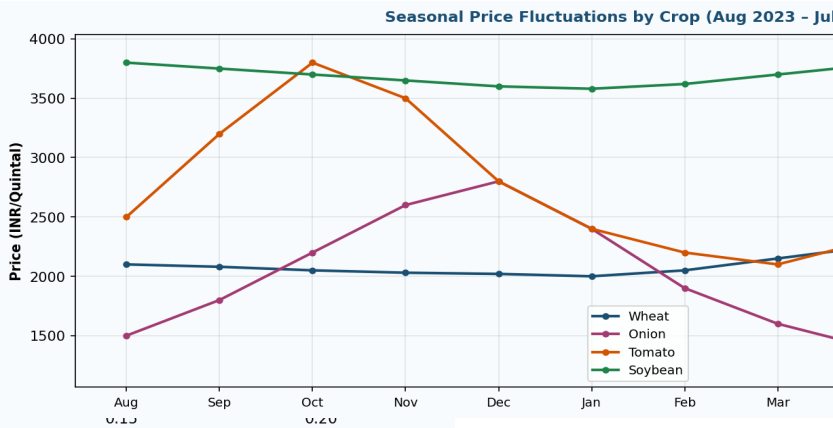
Figure 4 — Wheat Price Forecast: Actual vs. Predicted



The hybrid deep learning model closely tracks actual wheat prices over 90 days. The 30-day forward projection (blue dashed) includes a 95% confidence interval, demonstrating both accuracy and calibrated uncertainty quantification.

Figure 5 — Feature Importance Figure 6 — Seasonal Price Volatility





Left: Fuel price and crop arrival volume are the dominant price drivers. Right: Perishable vegetables (Onion, Tomato) exhibit extreme seasonal swings, making them the most challenging — and highest-value — forecasting targets.

05 | CONCLUSIONS & FUTURE DIRECTIONS

A New Era for Agricultural Economics

This study demonstrates that machine learning — particularly hybrid deep learning ensembles combining LSTM/GRU, GNNs, CNNs, and gradient-boosted trees — provides a transformative capability for crop price forecasting. Achievable accuracy of $R^2 > 0.93$ and MAPE < 1.5% far surpasses the classical baselines that practitioners have relied upon for decades.

Implication	Beneficiary	Action

30-day price outlook with <1.5% error	Farmers	Optimise planting, harvesting & selling schedules
Real-time market intelligence dashboards	Merchants & Traders	Reduce inventory risk; improve procurement timing
Scenario modelling under climate shocks	Policymakers	Design targeted interventions & price-support schemes
Generalised multi-region model	Research Community	Expand dataset scope; enhance interpretability (Theofilou et al., 2025)

Future Research Directions

- Enhance **model interpretability** using SHAP values and attention maps to build stakeholder trust in AI-driven forecasts (Theofilou et al., 2025).
- Develop **generalised cross-region models** that transfer knowledge across different geographies and crop varieties without full retraining.
- Integrate **satellite remote-sensing** and IoT sensor data for real-time soil moisture and crop health signals as additional predictive features.
- Explore **Transformer and Temporal Fusion Transformer** architectures for long-horizon multi-step forecasting beyond the current 30-day window.